Episode_3

BLAST, Multiple Sequence Alignment and Phylogeny of β-Lactamases

Overview and learning outcomes

You will start with ONE protein (TEM-1 β-lactamase, the enzyme that makes many bacteria resistant to ampicillin) and use it as bait to find relatives in the NCBI databases. You will then collect 15 to 20 β-lactamase sequences (TEM, SHV, CTX-M, KPC, OXA, AmpC and others), align them, and draw an evolutionary tree.

By the end you should be able to:

- Explain what BLAST does and choose the right BLAST program (blastn, blastp, blastx, tblastn, tblastx).
- Set up a search on the NCBI BLAST web page and explain every important option.
- Interpret E-value, bit score, percent identity, query cover and the alignment itself.
- Build a multiple sequence alignment (MSA) and recognize conserved catalytic motifs.
- Build and read a phylogenetic tree, including bootstrap values and rooting.

Software needed: only a web browser for Parts 3 to 10. For the tree (Part 12) you can use a web server (NGPhylogeny.fr or IQ-TREE web server) or free desktop software (MEGA, Jalview, FigTree).

1. Background you need first

1.1 Why β-lactamases?

β-lactam antibiotics (penicillins, cephalosporins, carbapenems) kill bacteria by blocking cell-wall synthesis. β-lactamases are enzymes that break the β-lactam ring and make the antibiotic useless. They are a major cause of antimicrobial resistance (AMR). They are an excellent teaching family because they are well studied, very diverse, and the family has clear sub-groups.

1.2 Ambler classes (the classification you will see in your tree)

Class	Mechanism	Examples	Notes
A	Serine (active-site Ser70)	TEM, SHV, CTX-M, KPC, GES, PER	TEM-1 belongs here
B	Metallo (needs Zn2+)	NDM, VIM, IMP	Evolutionarily unrelated to A, C, D
C	Serine	AmpC, CMY	Often chromosomal in Enterobacterales
D	Serine	OXA family	Includes OXA-48 and OXA-23

Classes A, C and D share a distant common ancestor (they look like each other, slightly). Class B is a different protein fold. This matters later: Class B sequences make a good OUTGROUP to root the tree.

1.3 Key terms

- Query: the sequence you give BLAST (TEM-1).

- Subject (hit): a database sequence that BLAST says is similar.
- Similarity: how alike two sequences are (a measurement).
- Homology: shared ancestry (a conclusion). Sequences are either homologous or not; they are never '60% homologous'. We infer homology from significant similarity.
- HSP (high-scoring segment pair): a matching region between query and subject.
- Identity: positions with exactly the same residue. Positives (similarity): identical OR chemically similar residues (for example I and V).
- Gap: an insertion or deletion introduced to line up sequences.
- Ortholog / paralog: the same gene in different species / a duplicated gene in the same lineage. TEM and SHV are close paralogs derived from a common ancestor gene.
- FASTA format: a text format. One line starting with > (the name), then the sequence.

2. The BLAST family: which program to choose?

BLAST (Basic Local Alignment Search Tool) compares your query with a database and reports local regions of similarity. Pick the program by asking: what is my query (DNA or protein) and what is the database (DNA or protein)?

Program	Query	Database	What happens	Typical use for this project
blastn	Nucleotide	Nucleotide	DNA vs DNA	Find blaTEM genes or plasmids carrying them; confirm a PCR product
blastp	Protein	Protein	Protein vs protein	MAIN tool here: TEM-1 protein vs protein databases
blastx	Nucleotide	Protein	Query translated in all 6 frames, each compared with protein DB	You have a gene or contig and do not know the protein; find which frame encodes the β-lactamase
tblastn	Protein	Nucleotide	Database translated in all 6 frames	Search unannotated genomes or contigs for a β-lactamase gene
tblastx	Nucleotide	Nucleotide	Both translated in 6 frames and compared	Very sensitive but slow; used for distant coding sequences; rarely needed

Why translate? DNA is less informative than protein. The genetic code is redundant, so different DNA can encode the same protein. Distant relatives often show clear similarity at protein level but almost none at DNA level. For evolutionary relationships among diverse β-lactamases, always use protein.

Other BLAST flavors on the same site: PSI-BLAST and DELTA-BLAST (more sensitive, for distant relatives; see Part 8), PHI-BLAST (pattern-seeded), and Align two or more sequences (compare your own sequences directly, no database).

3. Protocol A: Get the TEM-1 query sequence

1. Go to https://www.ncbi.nlm.nih.gov/protein/ (or UniProt, https://www.uniprot.org).
2. Search for: TEM-1 beta-lactamase. A well-known curated entry is UniProt P62593 (BLAT_ECOLX). Check the entry yourself: the protein should be about 286 amino acids long, beginning with a signal peptide (the start is MSIQHFRVALIPFFAAFCLPVFA...).
3. Open the entry and click FASTA. Copy the text (including the > line).
4. Save it as TEM1.fasta, or just keep it in your clipboard.

5. Optional DNA query for Part 7: search NCBI Nucleotide for blaTEM-1 (the classic source is the plasmid pBR322; check the record annotation and use the CDS only).

Tip: record the accession number of everything you use. Your report must be reproducible.

4. Protocol B: Run a BLASTp search step by step

1. Open https://blast.ncbi.nlm.nih.gov/Blast.cgi and click Protein BLAST (blastp).
2. Enter Query Sequence: paste the FASTA text, or type the accession (for example P62593). Job title: write something meaningful, such as TEM1_vs_nr.
3. Choose Search Set: Database = Non-redundant protein sequences (nr). Leave Organism blank at first.
4. Program Selection: Algorithm = blastp (protein-protein BLAST).
5. Leave Algorithm parameters at default for your first run (explained in Part 5).
6. Click BLAST. Wait for the results page (seconds to a minute).
7. Save your work: copy the RID (Request ID) at the top. It is valid for about 24 hours, and you can reopen it with Recent Results.

Repeat once with Database = RefSeq Select proteins or Swiss-Prot. Compare how the results change.

5. Every important option on the NCBI BLAST page explained

5.1 Enter Query Sequence

Option	Meaning
Query box	Paste FASTA or raw sequence, or type an accession or GI. Multiple FASTA entries are allowed (each is searched separately).
Upload file	Load a FASTA file instead of pasting.
Query subrange (From, To)	Search only part of the query, for example, only the mature protein after the signal peptide or only a domain.
Align two or more sequences	Turns the page into a direct pairwise or small multiple comparison with no database.
Job title	Your label for the search.

5.2 Choose Search Set

Option	Meaning
Database	nr (everything, redundant, huge); RefSeq Select proteins (one curated representative per species or group, smaller and cleaner); Reference proteins (RefSeq, non-redundant); Swiss-Prot (manually reviewed, small, high quality); PDB (proteins with 3D structures); Model organisms (landmark); Metagenomic and TSA databases. Available choices differ between blastp and blastn.
Organism	Restrict to a taxon (for example Escherichia coli, or Enterobacterales). You can add several organisms, or Exclude them.
Exclude: Models (XM/XP), Uncultured/environmental sample sequences	Removes predicted or uncultured sequences that are often poorly annotated. Useful for cleaner results.

Option	Meaning
Limit to: Sequences from type material	Only taxonomically reliable type-strain sequences.
Entrez Query	A text filter written like an NCBI search, for example: beta-lactamase[Title] NOT hypothetical[Title].

5.3 Program Selection

Option	When to use
Quick BLASTP (accelerated)	Fast search for close hits (high identity). Good first look.
blastp	Standard, more sensitive; default for homology work.
PSI-BLAST	Iterative: builds a profile from hits and searches again; finds distant relatives.
DELTA-BLAST	Uses a database of conserved domains to start a profile search; very good for distant family members.
PHI-BLAST	Finds hits that match your query AND a pattern (motif) you provide.

For blastn the options are Megablast (highly similar sequences, same species, fast), Discontiguous megablast (more dissimilar, good for cross-species coding sequences), and blastn (somewhat similar sequences; the classic, more sensitive than megablast for short or divergent queries).

5.4 Algorithm parameters (the 'Algorithm parameters' drop-down)

Parameter	Default (blastp)	Plain-language meaning and when to change it
Max target sequences	100	How many hits to keep. Increase (500 to 5000) if you want a bigger collection; decrease to simplify teaching. A low limit can hide some of the diversity you want.
Short queries	Automatically adjust for short queries	Leave on. Adjusts settings for peptides under about 30 residues.
Expect threshold (E)	0.05 (older versions 10)	Hits with an E-value above this are not shown. Lower it (1e-5, 1e-10) for stricter results; raise it (10, 100) when hunting very distant relatives.
Word size	5 (options 2, 3, 5, 6)	BLAST first looks for short exact word matches (seeds), then extends them. Smaller words are more sensitive but slower.
Max matches in a query range	0 (off)	Limits the number of hits that overlap the same region of the query. Useful for long queries.
Scoring matrix	BLOSUM62	Table scoring how good each amino acid substitution is. BLOSUM62 is good for general searches. BLOSUM80 or PAM30 suit very close sequences; BLOSUM45 or PAM250 suit distant ones.
Gap costs	Existence 11, Extension 1	Penalty for opening a gap (11) and for each extra gap position (1). Higher penalties produce fewer gaps.
Compositional adjustments	Conditional compositional score matrix adjustment	Corrects for biased amino-acid composition so that composition alone does not create false hits. Leave default.

Parameter	Default (blastp)	Plain-language meaning and when to change it
Filter: Low complexity regions	Off in blastp (on by default in blastn)	Masks repetitive or biased stretches (for example QQQQQQ) so they do not give false hits.
Mask for lookup table only	Off	Applies the filter only to seeding, not to the final alignment.
Mask lower case letters	Off	Lets you mask your own regions by writing them in lower case.

If you run blastn, you additionally see: Word size (28 for megablast, 11 for blastn), Match/Mismatch scores (for example 1,-2 for megablast; 2,-3 for blastn), and Gap costs. Strict scores speed up searches of very similar sequences.

5.5 After the search: formatting

On the results page use Reformat results or the Download menu. You can change the number of descriptions or alignments displayed, switch the alignment view (Pairwise, Query-anchored with identities, Flat query-anchored), and choose the output format (HTML, Text, XML, tab-delimited Hit table). The Filter results panel lets you restrict by percent identity, E-value or query cover, or by organism.

6. Reading the BLASTp results

The results page has these tabs: Descriptions, Graphic Summary, Alignments, Taxonomy and (for protein) Distance tree of results.

6.1 The Graphic Summary

A coloured bar chart where the top horizontal line is your query (positions along the bar) and each line below is a hit. Colour shows alignment score (bit score): below 40 black, 40 to 50 blue, 50 to 80 green, 80 to 200 magenta, 200 and above red. A red bar over the whole query is a strong full-length match. Gaps in a bar show regions without similarity.

6.2 The Descriptions table (the most important)

Column	What it tells you	How to use it
Description	Name of the hit	Names can be wrong or vague (for example 'class A beta-lactamase'). Do not trust the name blindly.
Scientific name	Organism	Check for horizontal gene transfer: the same gene appears in many species.
Max score	Bit score of the best HSP	Higher means better. Independent of database size.
Total score	Sum of bit scores of all HSPs	Differs from Max score if there are several segments.
Query cover	% of your query covered by the alignment	Distinguishes a full-length homolog (about 100%) from a shared domain only.
E-value	Number of hits with this score you would expect by chance in a database of this size	Smaller is better; 0.0 means essentially impossible by chance.
Per. ident	% identical amino acids in the aligned	Interpret with query cover, never alone.

Column	What it tells you	How to use it
	region	
Acc. Len	Length of the subject	Very different from query length suggests a fragment or domain fusion.
Accession	Database ID	Use this to download the sequence and for your report.

6.3 The numbers, explained

- Raw score: sum of substitution scores (from BLOSUM62) minus gap penalties.
- Bit score: the raw score normalised so that scores from different searches can be compared. Roughly, every +1 bit doubles the evidence.
- E-value: depends on bit score AND on database size (a bigger database gives more chance matches). Rough guide for proteins: below 1e-30 very strong homolog; 1e-10 to 1e-5 likely homolog; around 0.01 to 1 is a borderline hit; above 1 usually noise (but see Part 8).
- Percent identity guide for proteins: above 40% over most of the length almost always means homology and probably similar function; 25 to 40% is probably homologous (the 'twilight zone' near 25%, so confirm with structure or motifs); below 20% is unreliable without profile methods.
- Query cover: always check it. A 90% identical hit covering only 20% of the query is a small domain, not a true full-length homolog.
- Positives: includes conservative substitutions, so Positives is always at least as large as Identity.

6.4 What to expect for TEM-1

- The top hundreds of hits are 100% identical proteins labelled TEM-1 from many species and plasmids. This is redundancy, not diversity. The hit list also contains point mutants (TEM-2, TEM-3 and so on, mostly 99% identical, each differing by a few residues).
- Close relatives follow: SHV family (roughly 65 to 70% identity). Further down: CTX-M, KPC, PER, GES and other class A enzymes (roughly 35 to 45% identity, with good query cover).
- Class C (AmpC) and class D (OXA) enzymes are usually below 30% identity and often absent from a default top-100 list. Metallo-β-lactamases (class B) will not be found, because they are unrelated in sequence.

Practical consequence: a plain TEM-1 blastp search will not give you OXA, AmpC or NDM. You collect those either by searching their names or by using DELTA-BLAST/PSI-BLAST (Parts 8 and 9).

6.5 Reading one alignment

Click the Alignments tab. Each block shows Query, a middle line, and Sbjct, with residue numbers. In the middle line a letter means identical residue, a plus (+) means a similar residue, and a blank means a mismatch. Dashes in one sequence are gaps. Look for the conserved motif S-T-F-K (Ser70 in TEM-1) in your hits. If the motif is intact in a hit, the enzyme probably still has the serine active site.

7. Protocol C: nucleotide programs in practice

Use these small exercises to see how the programs differ.

7.1 blastn (DNA vs DNA)

1. Open Nucleotide BLAST. Paste the blaTEM-1 CDS (nucleotide FASTA).

2. Database: Nucleotide collection (nr/nt). Program: Megablast first (highly similar). Click BLAST.
3. Expect hundreds of 100% hits on plasmids from many species. This shows that the gene is mobile (horizontal gene transfer).
4. Re-run with Discontiguous megablast, then blastn. Compare how many hits you get at low identity. Which one finds more distant relatives?
5. Interpretation tip: for nucleotide hits, above 95% identity usually means the same gene (or allele); below 70% is hard to align reliably.

7.2 blastx (DNA query, protein database)

1. Open blastx. Paste the same DNA (or a longer plasmid or contig segment containing it).
2. Database: nr. Run it.
3. Look at Frame in the alignment (for example +1, +2, -1). BLAST tried all 6 reading frames and found the one that encodes a β-lactamase.
4. When to use it: sequencing results from an unknown insert, a contig of unknown function, or checking where a start codon lies.

7.3 tblastn (protein query, translated DNA database)

1. Open tblastn. Paste the TEM-1 protein.
2. Database: choose Whole-genome shotgun contigs (WGS) and restrict Organism to a species of interest (for example Klebsiella pneumoniae), or use Genomic plus transcript databases.
3. Run. Hits show where in an unannotated genome a β-lactamase gene may hide, even if the gene was never annotated.

7.4 tblastx (translated DNA vs translated DNA)

Use only when you need the most sensitive coding-level comparison between two DNA sets. It is computationally heavy: 36 frame comparisons per hit. Notice how few options remain on the page.

8. Protocol D: find distant relatives (DELTA-BLAST or PSI-BLAST)

1. On the protein BLAST page keep the TEM-1 query and choose Algorithm = DELTA-BLAST (database: nr or Reference proteins).
2. Set Max target sequences to 500 to see more of the family. Run.
3. For PSI-BLAST: choose PSI-BLAST, run, then on the results page click Run PSI-BLAST iteration 2. Tick or untick the hits to include in the profile (remove obviously unrelated ones to avoid profile corruption). Repeat for iterations 3 and 4 until no new sequences appear (convergence).
4. Compare with the plain blastp results: now OXA, AmpC and other classes should appear, with weaker E-values (1e-5 to 1e-20).

Warning: iterative searches can drift. If a false hit enters the profile, later iterations bring in unrelated proteins. Always inspect which hits you accept.

9. Protocol E: choose 15 to 20 sequences for the alignment

Goal: a balanced set that covers all Ambler classes without including 15 copies of the same protein.

9.1 Suggested panel (20 sequences)

#	Enzyme	Ambler class	Why include it
1	TEM-1	A	Your query
2	TEM-52 (an ESBL variant) or another TEM variant	A	Shows how close variants cluster
3	SHV-1	A	Nearest paralog family to TEM
4	SHV-12	A	ESBL variant of SHV
5	CTX-M-15	A	Most widespread ESBL
6	CTX-M-14	A	Another CTX-M group
7	KPC-2	A	Carbapenemase
8	GES-1	A	Class A ESBL, divergent
9	PER-1	A	Divergent class A ESBL
10	BlaZ (Staphylococcus aureus penicillinase)	A	Gram-positive class A
11	AmpC (Escherichia coli)	C	Chromosomal cephalosporinase
12	CMY-2	C	Plasmid-borne AmpC
13	OXA-10	D	Classic OXA ESBL-type
14	OXA-48	D	Carbapenem-hydrolysing OXA
15	OXA-23	D	Acinetobacter carbapenemase
16	OXA-51 (intrinsic in A. baumannii)	D	Chromosomal OXA
17	NDM-1	B	Metallo-β-lactamase, outgroup
18	VIM-2	B	Metallo-β-lactamase, outgroup
19	IMP-1 (optional)	B	Third MBL, for a stronger outgroup
20	A non-β-lactamase penicillin-binding protein fragment (optional)	none	Only if you want an advanced distant-group demonstration

You can cut this to 15 by removing items 2, 4, 16, 19 and 20. Class names are used for naming only; always confirm the class by reading the literature for each accession.

9.2 How to get each sequence

1. From your DELTA-BLAST or blastp results, tick the boxes next to chosen hits. This works for TEM, SHV, CTX-M, KPC, GES and PER.
2. For enzymes not in the hit list (AmpC, OXA, NDM, VIM, IMP): go to NCBI Protein and search the enzyme name, for example OXA-48 beta-lactamase. Restrict Source database to RefSeq or use UniProt reviewed (Swiss-Prot) entries. Prefer entries whose title names the exact variant. NCBI's Reference Gene Catalog / AMRFinderPlus pages are good places to confirm exact variant names.
3. Select the records, click Send to, then Complete Record, File, Format FASTA, and Create file.
4. Combine all records into one text file called betalactamases.fasta.
5. Rename the headers to short unique names, for example >TEM-1_Ecoli, >SHV-1_Kpneu, >CTXM15_Ecoli, >KPC2_Kpneu, >OXA48_Kpneu, >NDM1_Kpneu. Short names with no spaces, brackets or colons avoid errors in alignment and tree programs.
6. Check: count the > symbols (should be 15 to 20) and that no sequence appears twice.

Selection tips: choose sequences from different species when possible, avoid identical sequences (they add no information), and avoid fragments (partial sequences cause gap-filled alignments). Decide whether to include the signal peptide; keeping full-length is fine, and trimming to the mature protein is an optional refinement.

10. Protocol F: build the multiple sequence alignment (MSA)

An MSA lines up homologous positions in many sequences so that each column represents one position that evolved from the same ancestral residue. It is the foundation of a good tree: a bad alignment means a bad tree.

10.1 Using Clustal Omega on the EMBL-EBI web server

1. Go to https://www.ebi.ac.uk/jdispatcher/msa/clustalo (EBI Job Dispatcher; Clustal Omega).
2. STEP 1: Choose Protein as sequence type. Paste your multi-FASTA or upload betalactamases.fasta.
3. STEP 2: Output format: ClustalW with character counts (default) is fine. You can also choose Pearson/FASTA for later steps. Leave the other options at default for the first run (Dealign input sequences = No; Mbed-like clustering guide-tree = Yes; Iterations = 0).
4. Enter an email (optional) and click Submit.
5. On the results page see: Alignments (text), Result Summary, Guide Tree and Phylogenetic Tree (Clustal Omega's quick tree), and the Percent Identity Matrix, which lists pairwise identity for every pair of sequences. Save the alignment file and the identity matrix.

Alternatives: MUSCLE (EBI), MAFFT (https://mafft.cbrc.jp/alignment/server/, very good for divergent sequences; try option L-INS-i for accuracy with 20 sequences), and COBALT (on NCBI BLAST results, tick hits and click Multiple alignment).

10.2 View and edit with Jalview (free desktop software) or the EBI viewer

1. Open Jalview and load the alignment (File, Input Alignment).
2. Colour, Clustalx (colours by residue type) or Colour, Percentage Identity.
3. Use Calculate, Sort, by Tree order to place related sequences near each other.
4. Remove clearly misaligned or very gappy sequences only if justified. Trimming of poorly aligned columns can be done in Jalview (select columns, Edit, Remove columns) or with trimAl (automated).

11. Reading the alignment

- Conserved columns (asterisks or dark colour) are positions that evolution kept almost unchanged: they are often functionally important.
- Strongly vs weakly similar columns are shown as colon (:) and period (.) in the Clustal text output.
- Look for these classic catalytic regions (residue numbers refer to TEM-1; sequence positions in other enzymes differ because of gaps, which is why scientists use the standard Ambler numbering):
  - Class A: S70-x-x-K73 (S-T-F-K in TEM-1), S130-D-N132, the Ω-loop with Glu166, and K234-(T/S)-G236 with Arg244.
  - Class C and D: they have an S-x-x-K motif too, in a different position in the alignment, and their other motifs differ from class A.
  - Class B (NDM, VIM, IMP): H-x-H-x-D motif binds zinc. The metallo-enzymes will NOT line up with the serine enzymes at the active site, which is the expected result.
- Use the percent identity matrix. Typical values: TEM versus SHV about 68%, class A versus class C or D often 15 to 25%, class A versus class B below 15%. A 15% alignment is unreliable; do not interpret

the exact alignment of those regions with confidence.
- Insertions and deletions (gaps) are usually in surface loops; conserved cores are interrupted by few gaps.
- A sequence with very many gaps usually means a fragment, wrong sequence, or an unrelated protein.

Mini exercise: find the S-x-x-K motif in each sequence and write a table of its exact residues. Which sequences deviate?

12. Protocol G: construct the phylogenetic tree

A phylogenetic tree is a hypothesis about how the sequences diverged from common ancestors. Tips are sequences you sampled, internal nodes are inferred ancestors, branch length shows the amount of change.

12.1 Choose a method (and why)

Method	Idea	Pros	Cons	Use for
Neighbor-Joining (NJ)	Builds a tree from a pairwise distance matrix	Very fast, simple, good for learning	Less accurate for distant sequences	Quick first tree
Maximum Likelihood (ML)	Finds the tree that makes the data most probable under an evolutionary model	Accurate, standard in publications	Slower, needs a model	Your final tree
Bayesian	Samples trees by probability	Gives posterior probabilities	More complex	Advanced work

For this practical use NJ first (to learn the idea), then ML (final).

12.2 Option 1: NGPhylogeny.fr (web, easiest)

1. Open https://ngphylogeny.fr and choose a workflow such as 'PhyML+SMS' (ML with automatic model selection) or 'FastME' (distance-based). Both work with a simple upload.
2. Upload your alignment (FASTA or Clustal). In the workflow, enable alignment curation (BMGE or Gblocks) to remove poorly aligned columns; leave default settings.
3. Run. The result page shows a tree (and the model chosen). Download the Newick file.
4. View and edit it in iTOL (https://itol.embl.de) or FigTree.

12.3 Option 2: IQ-TREE web server

1. Open the IQ-TREE web server (iqtree.cibiv.univie.ac.at; check it is still available).
2. Upload the alignment. Choose Auto (ModelFinder) for the substitution model. Enable Ultrafast bootstrap with 1000 replicates.
3. Run and download the tree (treefile, Newick) and the log (shows the chosen model, for example LG+G4).

12.4 Option 3: MEGA (free desktop program)

1. Open MEGA, then Align, Edit/Build Alignment, Retrieve from file (or use File, Open A File). You can run Align by MUSCLE inside MEGA.
2. Phylogeny, Construct/Test Neighbor-Joining Tree (or Maximum Likelihood Tree).

3. Settings: Test of Phylogeny = Bootstrap method, 500 to 1000 replicates. Substitutions type = Amino acid. Model: for NJ use Poisson or JTT; for ML select Phylogeny, Find best DNA/Protein models first, then use the top-ranked model (often LG+G or JTT+G). Rates among sites: Gamma Distributed (G) with 4 discrete categories. Gaps/Missing data treatment: Partial deletion (site coverage cutoff about 90 to 95%) or Pairwise deletion.
4. Compute. The tree appears in the Tree Explorer. Use Root on Branch or Subtree/Root to set the root (see Part 13).
5. Export the tree as Newick (for FigTree and iTOL) and as an image.

12.5 Rooting the tree

Unrooted trees show relatedness but not direction in time. To root, choose an OUTGROUP: a sequence known to be more distant from all others than they are from one another. Here use NDM-1, VIM-2 (and IMP-1 if used), because class B is unrelated to the serine enzymes. In FigTree select the outgroup branch and click Reroot; in iTOL click the branch and select Set as outgroup (Tree structure options).

13. Reading the tree

- Topology: who groups with whom. Expect clades roughly matching the Ambler classes: class A (with TEM and SHV neighbours, CTX-M, KPC, GES and PER as separate branches), class C, class D, and class B as the outgroup.
- Branch length: horizontal distance proportional to the number of amino acid changes per site (check the scale bar). In a cladogram (no lengths) branch length is meaningless; make sure your picture is a phylogram.
- Bootstrap support: percent of re-sampled trees containing that branch. Roughly: 95 to 100 strongly supported; 70 to 95 reasonably supported; under 50 weak (collapse such nodes mentally). For IQ-TREE ultrafast bootstrap use 95 or more as strong.
- Sister groups: TEM and SHV should be close; CTX-M and KPC should also lie within class A but form separate lineages. If a sequence sits in the wrong class, check the sequence name and accession (mislabelling happens), or whether it is a fragment.
- Do not read the tip order from top to bottom as evolutionary time. Branches can be rotated around any node without changing the tree.
- Long branches may attract each other (long-branch attraction) and may give false groupings; check by removing one and re-running.
- Interpret honestly: a protein tree of distant families (below 25% identity) has poorly resolved deep branches. The tree is good for showing the families, not the exact order of the oldest divergences.

Write a short interpretation paragraph: which enzymes are most similar, which classes form clades, what the support values say, and one limitation.

14. Common mistakes and troubleshooting

Problem	Likely reason	Fix
Only identical TEM hits appear	nr is redundant and top-100 is full of copies	Use RefSeq Select or Swiss-Prot; filter by Per. ident below 99; raise Max target sequences
No OXA, AmpC or NDM in results	Too distant for plain blastp	DELTA-BLAST or PSI-BLAST, or search by name in NCBI Protein
Hit says 'hypothetical protein'	Poor annotation	Check domain architecture (Conserved Domains section) and the motif

Problem	Likely reason	Fix
Alignment looks messy	Mixed classes, fragments or duplicates	Remove fragments and duplicates; try MAFFT L-INS-i; trim columns
Tree program says duplicate names	Headers repeated or contain spaces	Give each sequence a unique short name
Tree has very low bootstrap	Too divergent or too short alignment	Trim gappy columns; use ML; limit to one or two classes
Percent identity seems wrong	Query cover is partial	Read identity together with query cover
RID expired	Results are kept about 24 hours	Download hit table and alignments soon

15. Student report checklist and review questions

Report checklist:

- Query accession and date of search, BLAST program, database, and every non-default setting.
- Screenshot of the Graphic Summary and a table of the 15 to 20 chosen sequences with accession, species, class, and percent identity to TEM-1.
- The alignment (screenshot of the conserved motifs) and the identity matrix.
- The tree (rooted, with bootstrap values) and a written interpretation.

Review questions:

1. Why is a protein BLAST usually better than a DNA BLAST for finding distant relatives?
2. A hit has E-value 1e-4, 28% identity and 30% query cover. Is it a convincing homolog of the full TEM-1? Why or why not?
3. What does the E-value change if the database becomes ten times larger?
4. When would you use blastx rather than blastp? When tblastn?
5. Why can we not simply align NDM-1 with TEM-1 and trust the catalytic site alignment?
6. What does a bootstrap value of 62 on a node mean, and how should it influence your conclusions?
7. Why do we root the tree with class B enzymes?

Good luck, and remember: BLAST gives evidence of similarity; the biological conclusion (homology and function) is your inference, and it should always be checked with motifs, structure or the literature.